



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 16: Long-run equilibrium

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ICS2523 Microeconomics

Lecture 16

Last lecture we

- Talk about the profit function
- Learn how to estimate the producer's surplus (in the short run)
- Learn the Partial Equilibrium Competitive Model

In today's lecture, we will finish **Competitive Markets**. We will

- Do a comparative statics analysis of the long-run equilibrium
- Determine the producer surplus in the long-run
- Welfare analysis in the long-run
- Recall some behavioural assumptions

Questions from last lecture

- How are the profits related to economies of scale? (since profits are positive the moment $MC > AC$).
- **Review Encuesta de Medio Semestre**

The Partial Equilibrium Competitive Model

Competitive Markets

1. The Partial Equilibrium Competitive Model (Review)

What is a **perfectly competitive industry**?

1. There is a large number of firms, each producing the same homogeneous product.
2. Each firm attempts to maximise profits
3. Each consumer attempts to maximise his utility
4. Each firm is a price-taker
5. Prices are assumed to be known by all market participants (information is perfect)
6. Transactions are costless
7. Free entry and exit of firms (only for the long-run)

1. The Partial Equilibrium Competitive Model (Review)

<http://www.youtube.com/watch?v=9Hxy-TuX9fs>

Remember individual's demand? (Marshallian)

quantity of x demanded = $x(P_x, P_y, I)$

$$\text{Market demand for } X = \sum_{i=1}^n x_i(P_x, P_y, I_i)$$

How is this curve (market demand) done graphically?

→ HORIZONTAL SUM

Timing of the Supply Response and Pricing

Competitive Markets

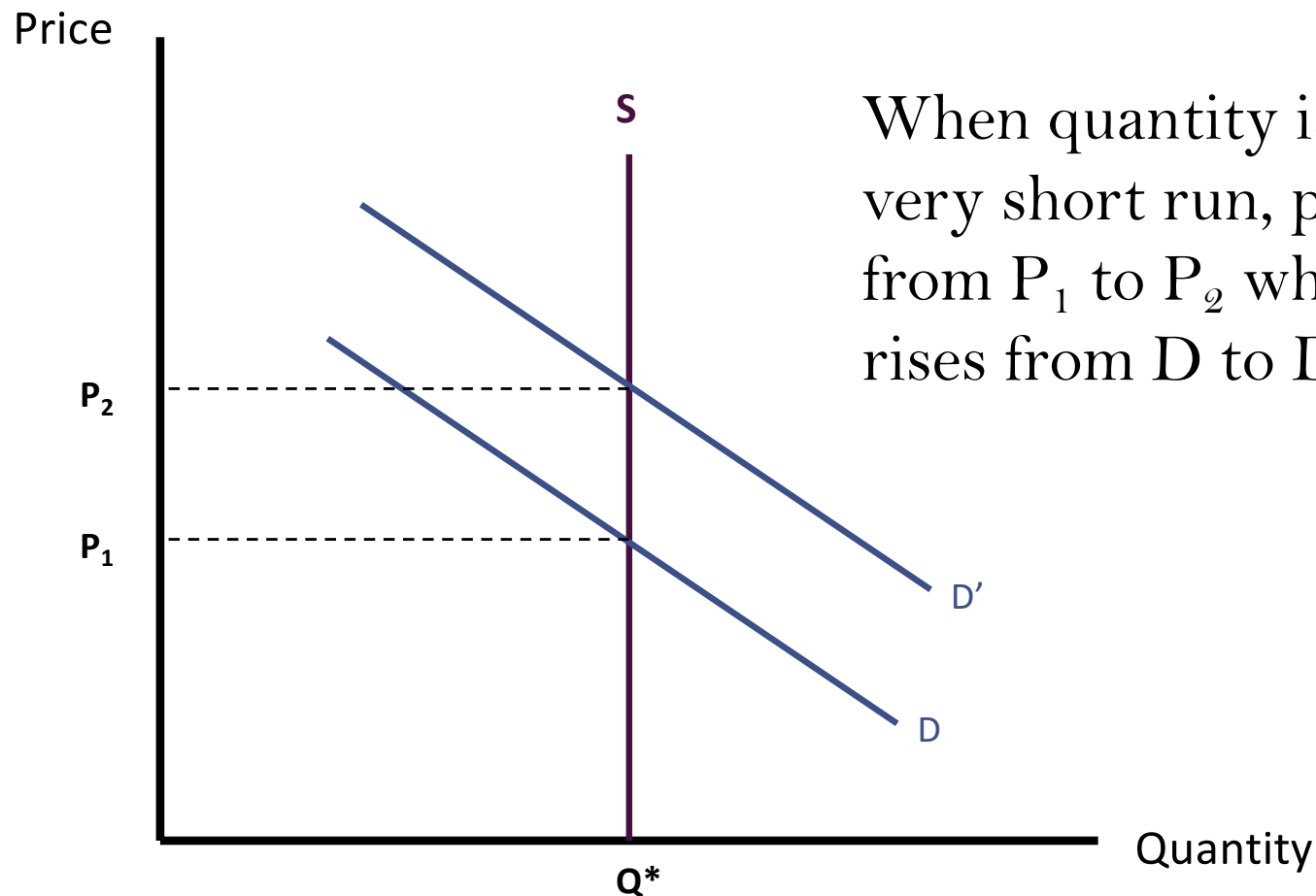
2. Timing of Supply Response (Review)

In the competitive pricing analysis, it is important to decide the length of time to be allowed for a supply response to changing demand conditions.

- Pricing in the **very short-run** → no supply response
→ Supply is fixed at Q^*
- Pricing in the **short-run** → number of firms in an industry is fixed (they can adjust the quantity they are producing)
- Pricing in the **long-run** → new firms may enter an industry

3. Pricing in the very short-run (Review)

Pricing in the very short-run



3. Pricing in the short-run

(Review)

Short-run market supply curve

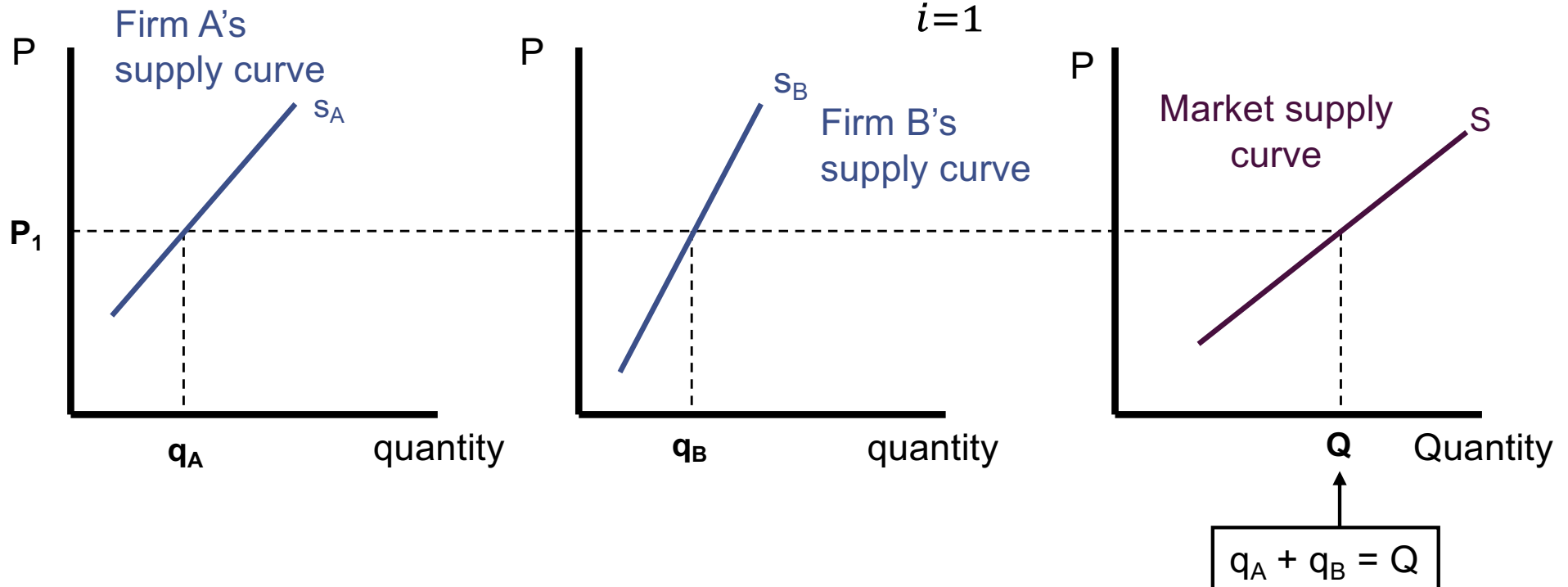
- The quantity of output supplied to the entire market in the short-run is the sum of the quantities supplied by each firm.
- This quantity depends on the price.
- Since each firm's supply curve has a positive slope, the market supply curve will also have a positive slope (Why?)
- Short-run market supply function:

$$Q_S(P, r, w) = \sum_{i=1}^n q_i(P, r, w)$$

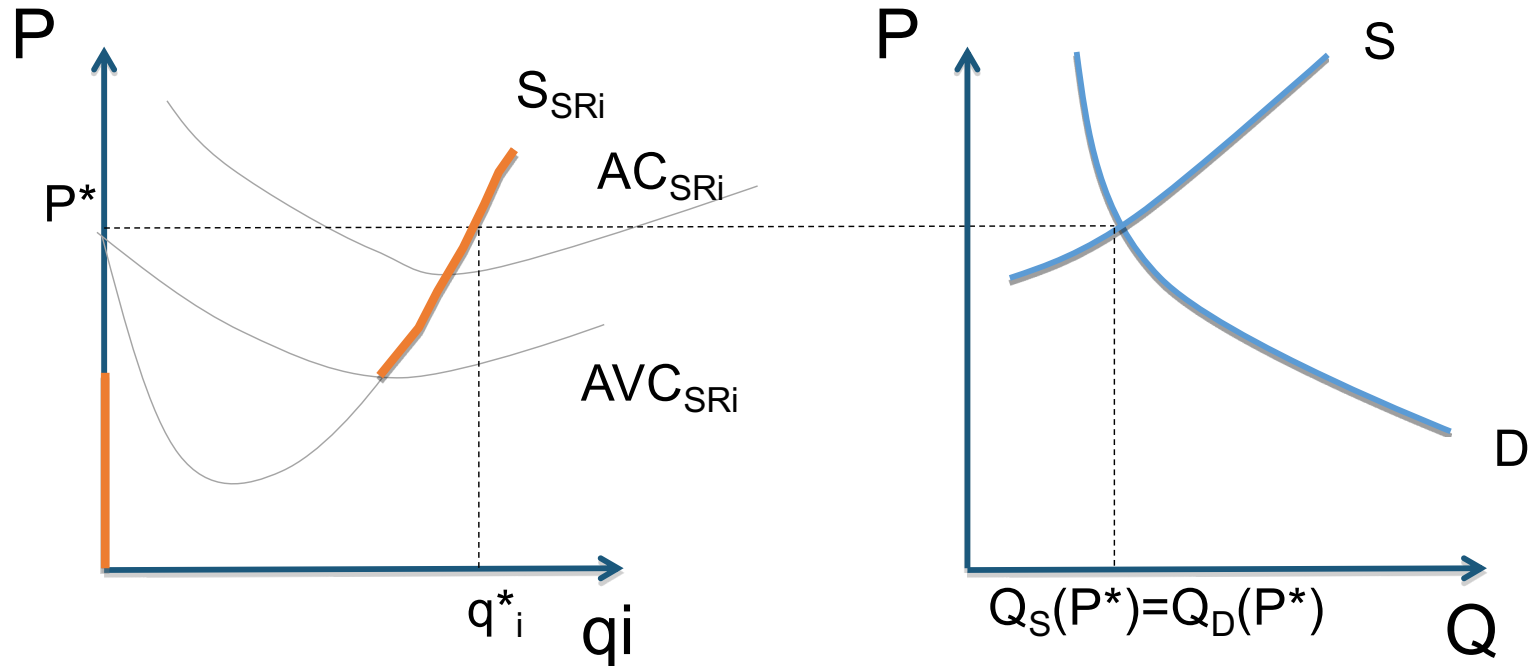
3. Pricing in the short-run (Review)

Short-run market supply curve

$$Q_S(P, r, w) = \sum_{i=1}^n q_i(P, k, w)$$



3. Pricing in the short-run (Review)



An equilibrium price (P^*) is one at which the quantity demanded is equal to the quantity supplied $\rightarrow$ at this price no one has incentives to alter their economic decisions

$$Q_D(P^*, P', I) = Q_S(P^*, r, w)$$
$$Q_D(P^*) = Q_S(P^*)$$

3. Pricing in the short-run

(Review)

Short-run supply elasticity

$$e_{S,P} = \frac{\partial Q_S}{\partial P} \cdot \frac{P}{Q_S}$$

Please note

- 1) Supply elasticity is POSITIVE → Why?
- 2) Relationship between elasticity and MC:
 - Low values of elasticity → It takes relatively large changes in price to induce firms to change their output levels → MC increase rapidly
 - High values of elasticity → Small increases in market price lead to a relatively large supply response by firms → MC do not rise steeply

3. Pricing in the short-run

(Review)

Shifts in the supply curve (short-run)

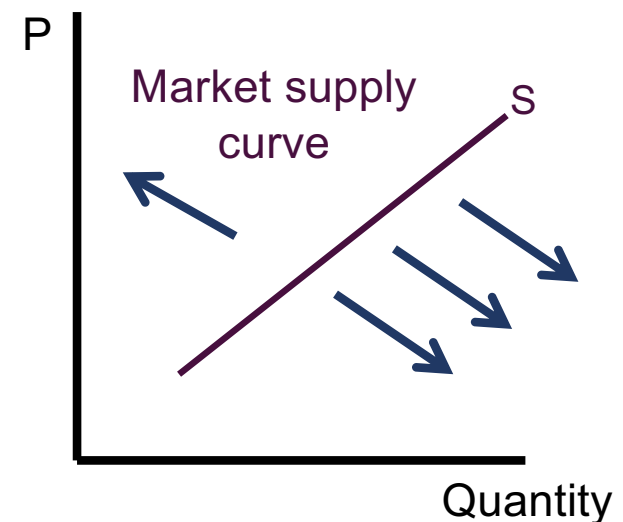
What happens to the supply curve in the short-run if...

The number of fixed inputs increases?

The price of the variable inputs increase?

There is a technological change?

Number of producers increase?



Activity 1

Which of the following changes will shift the SR supply curve for leather shoes?

- a) Increase in the rent of the shoe factory, which is fixed for 2 years
- b) Decrease in the price of leather, which is bought depending on the level of production
- c) Increase in the price of leather shoes
- d) Decrease in the price of boots



Wooclap code: **QPCEPM**

<https://app.wooclap.com/QPCEPM?from=even-t-page>

Long-run Equilibrium

Producer Theory

4. Long-run Equilibrium

We will use the **long-run costs curves**.

We will allow the **number of firms (n) to vary** in response to economic incentives $\rightarrow$ until the industry reaches an equilibrium (until no firm contemplating entry would be able to earn a profit).

We shall initially assume that all the firms in an industry have identical cost curves $\rightarrow$ no firm controls any special resources or technologies.

4. Long-run Equilibrium

Equilibrium conditions: a perfectly competitive industry is in a **long-run equilibrium** if there are **no incentives for profit-maximizing firms to enter or leave the industry**.

This will occur when:

1. The number of firms is such that $P=MC=AC$
2. Each firm operates at the low point of its long-run average cost curve

4. Long-run Equilibrium

Attention

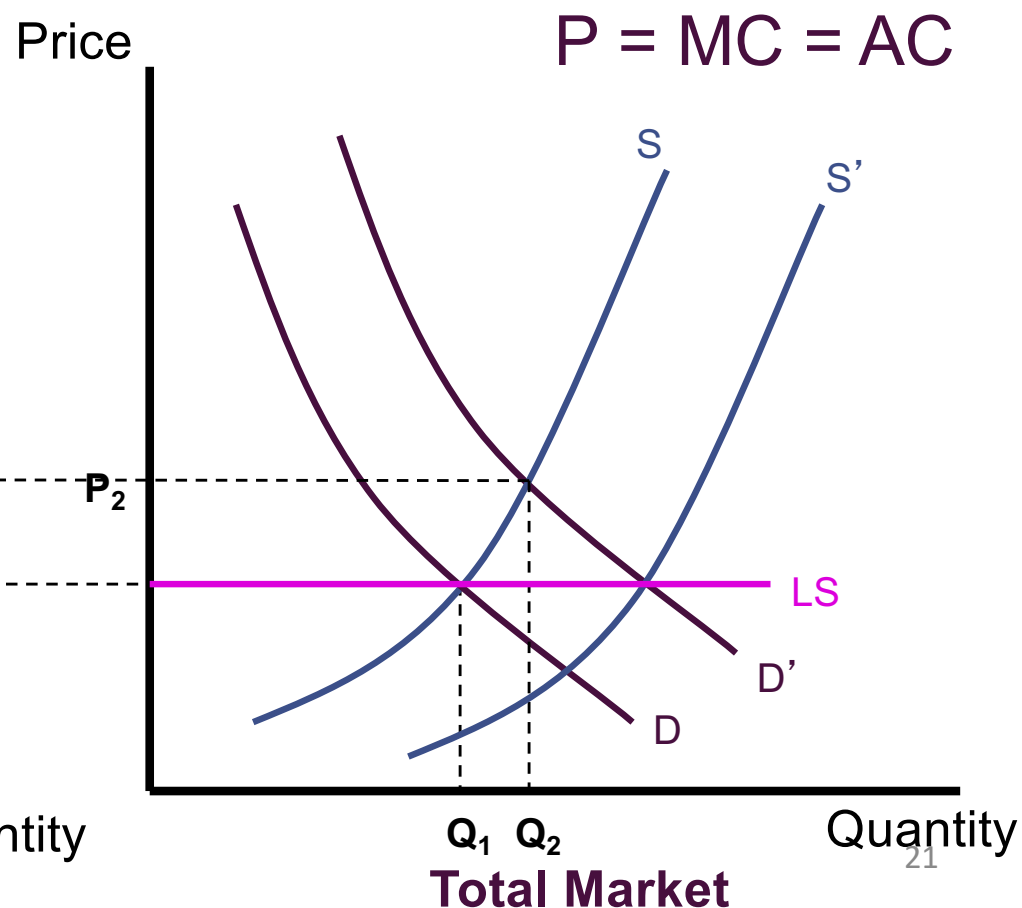
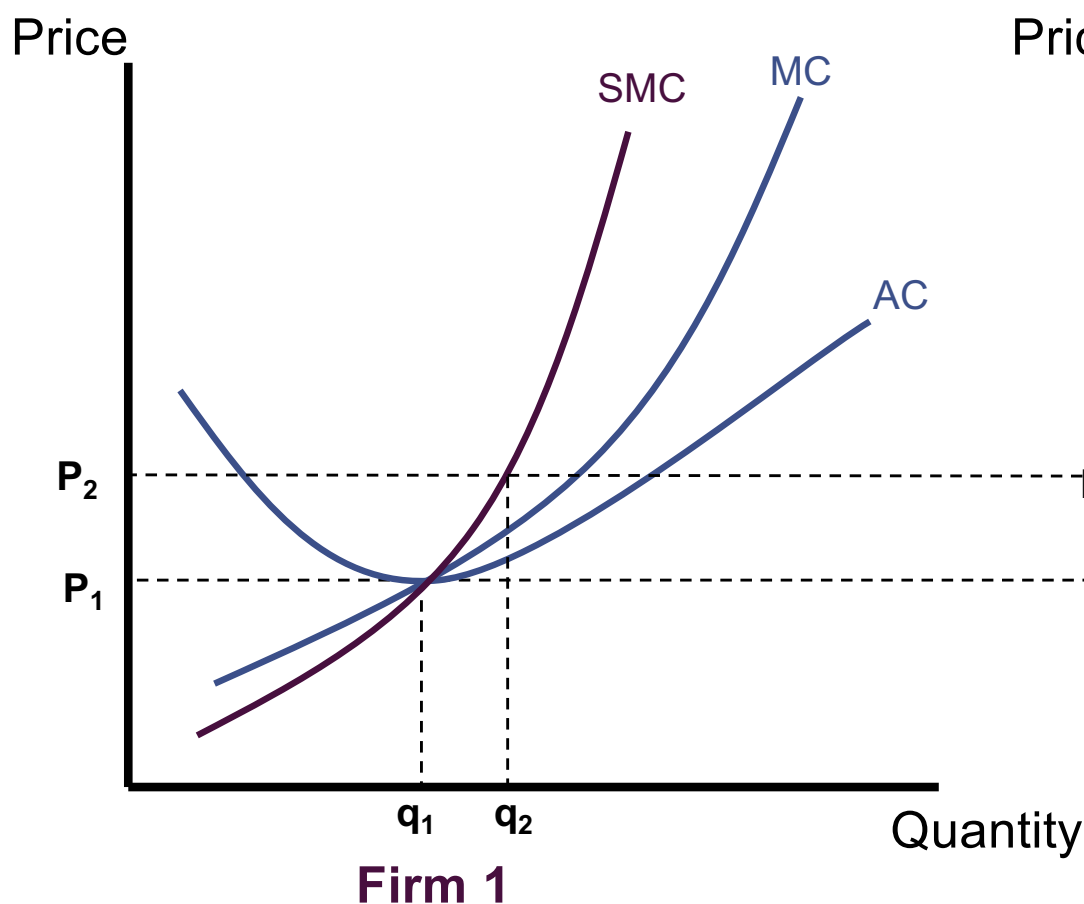
$P=MC=AC$ implies two important things that happen simultaneously:

1. $P=MC \rightarrow$ profit maximisation
2. $P=AC \rightarrow \pi=0 \rightarrow$ no incentives to enter or exit the industry

We will have to assume if the entry/exit of firms in the market affects the costs of the firms in the market.

4. Long-run Equilibrium: Constant Cost Industry

Assumption: entry of new firms into the industry has no effect on the prices of the firm's inputs $\rightarrow$ each firm will retain the same set of cost curves with which it started



Activity 2

Imagine a firm producing bicycle frames has the following long-run monthly costs:

$$C(q) = q^3 - 20q^2 + 100q + 8000$$

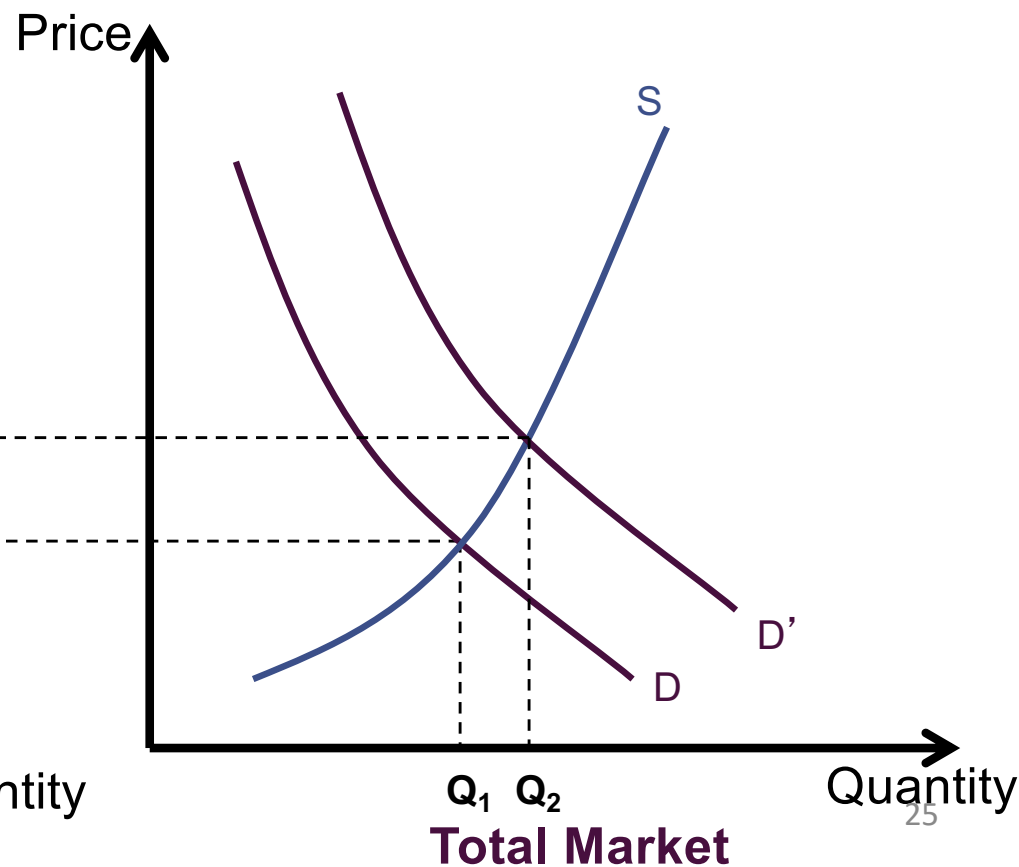
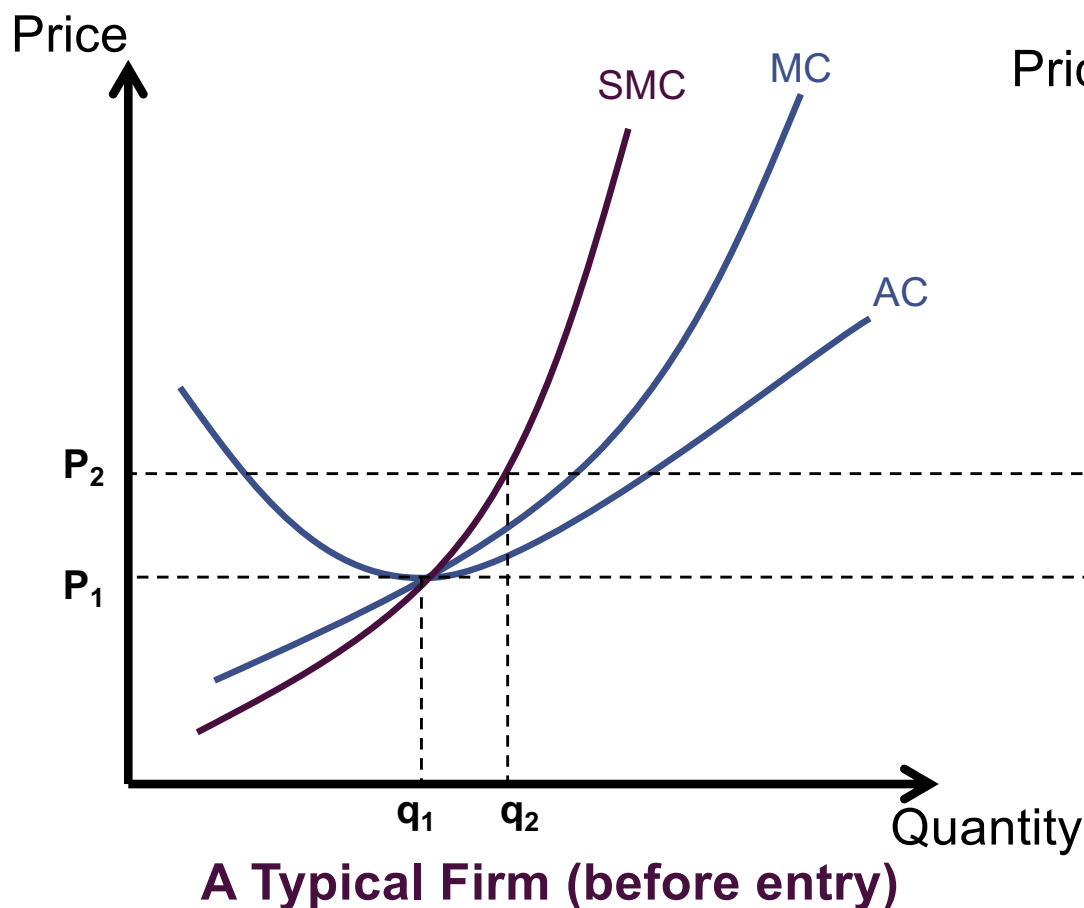
Where q is the number of frames produced per month, demand for the bicycle frames is given by:

$$Q_D = 2500 - 3P$$

- a) Determine the long-run equilibrium in this market (P^* , Q^* , q_i)
- b) How many firms are in the market?

4. Long-run Equilibrium: Increasing Cost Industry

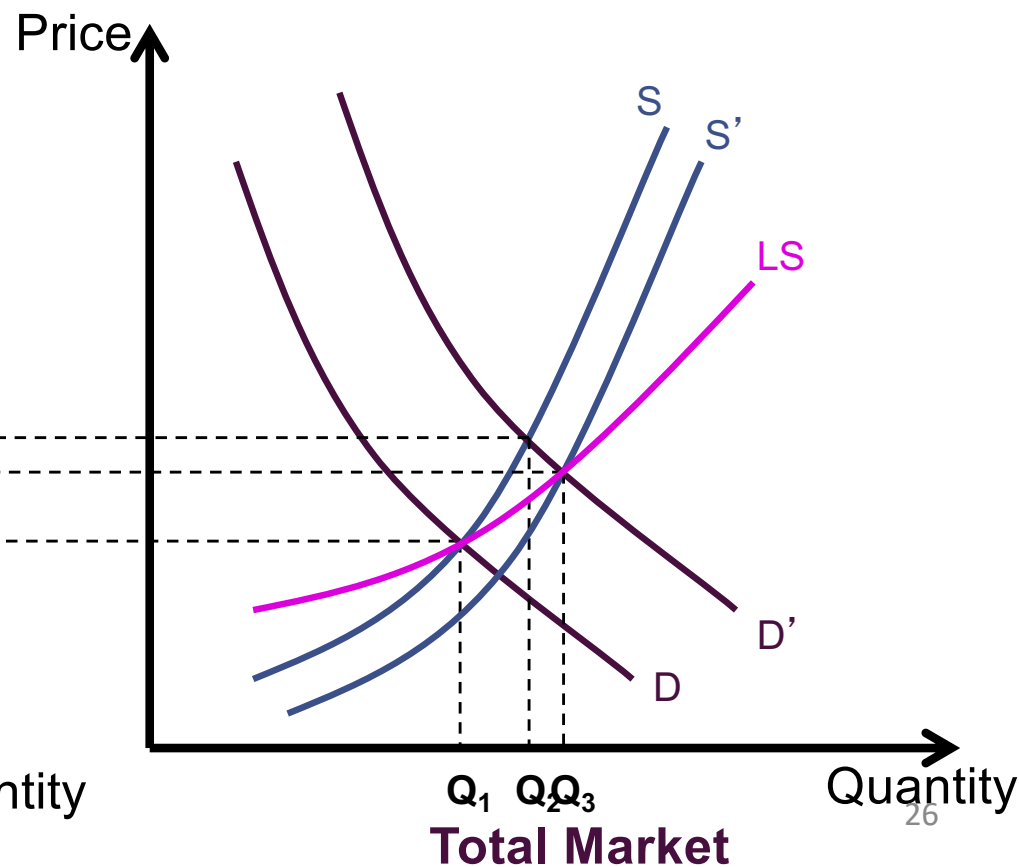
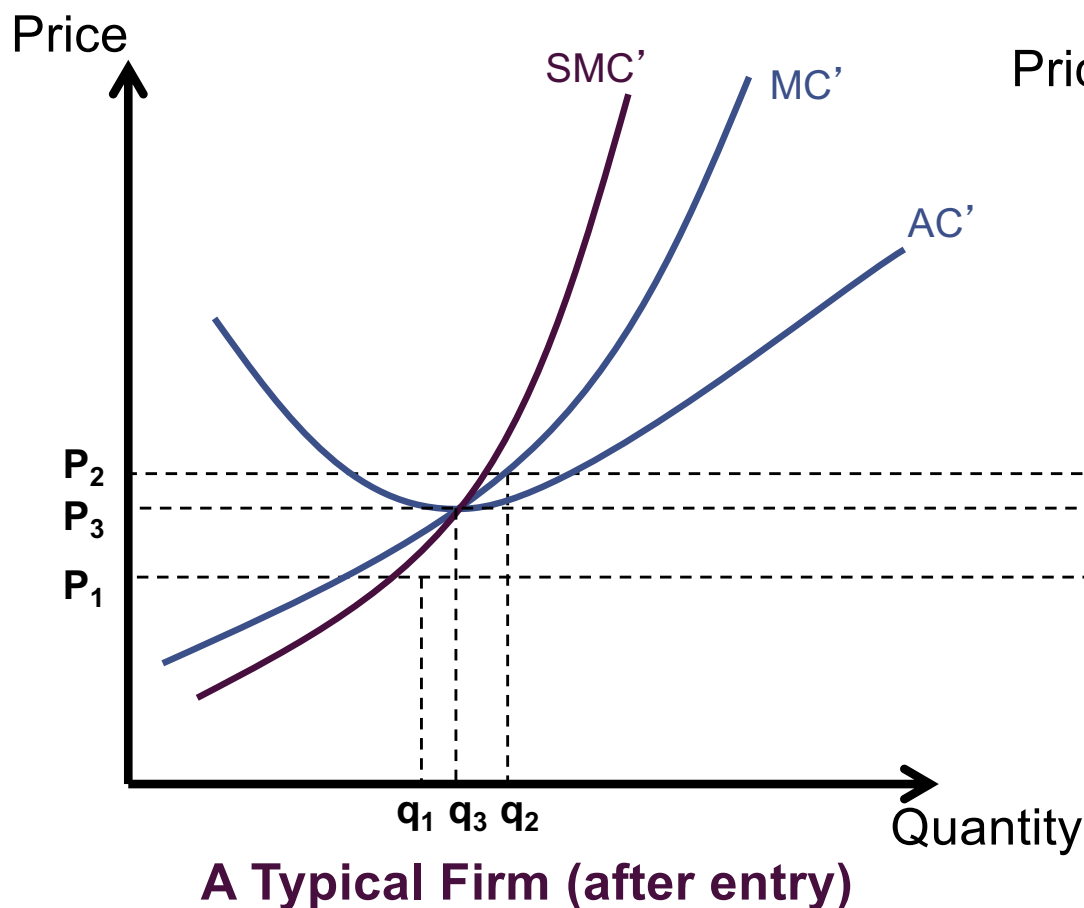
What happens if the entry of new firms causes the average costs of all firms to rise?



4. Long-run Equilibrium: Increasing Cost Industry

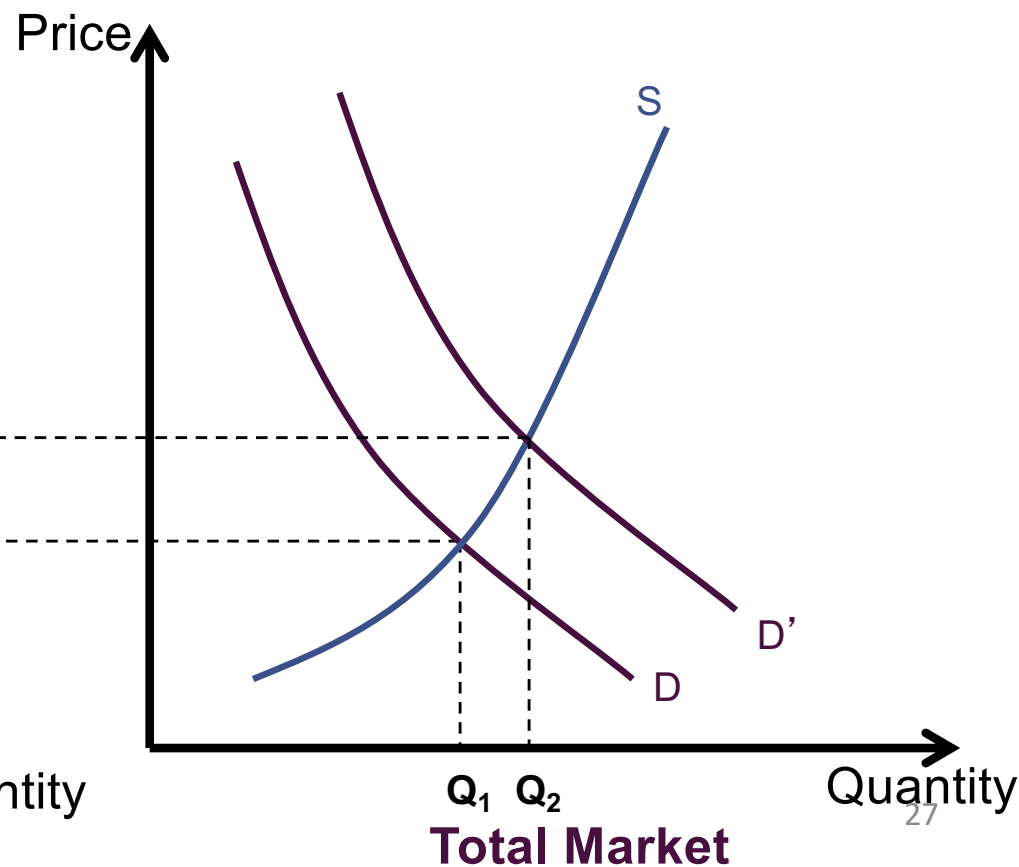
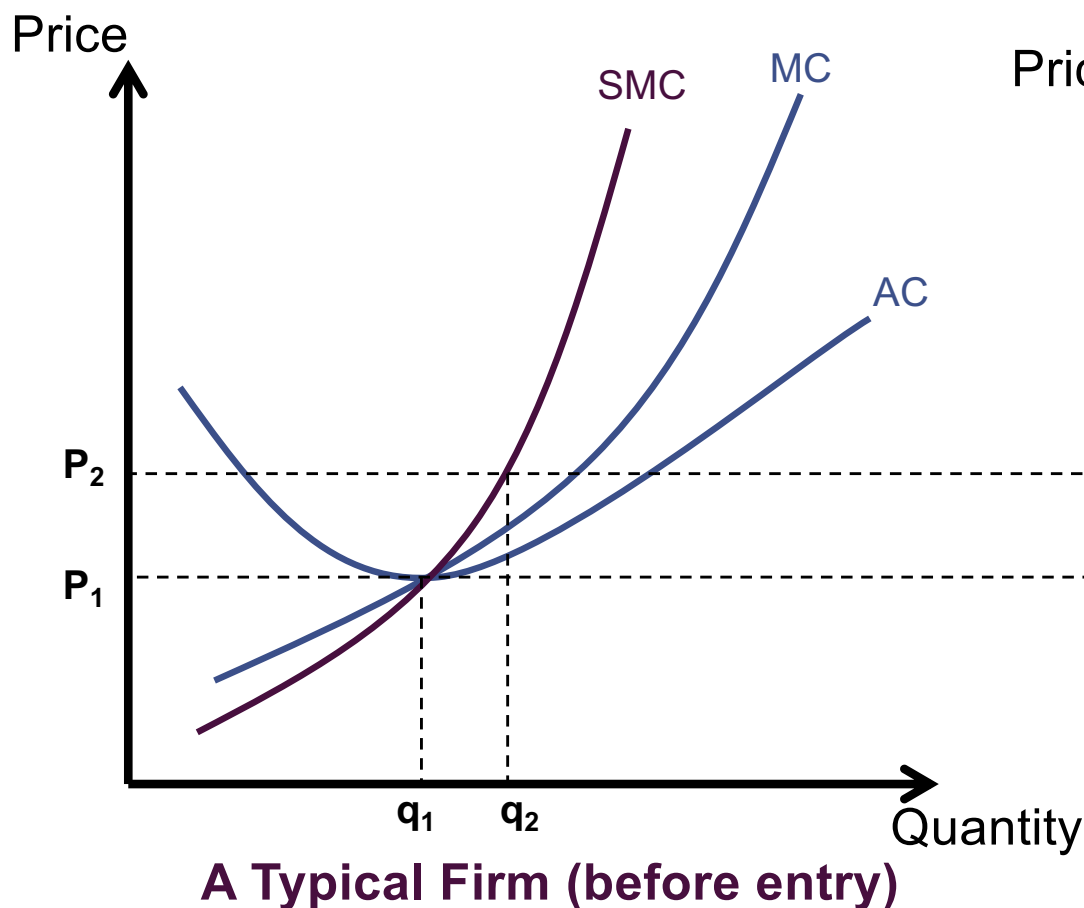
What happens if the entry of new firms causes the average costs of all firms to rise?

The long-run supply curve will be upward-sloping!



4. Long-run Equilibrium: Decreasing Cost Industry

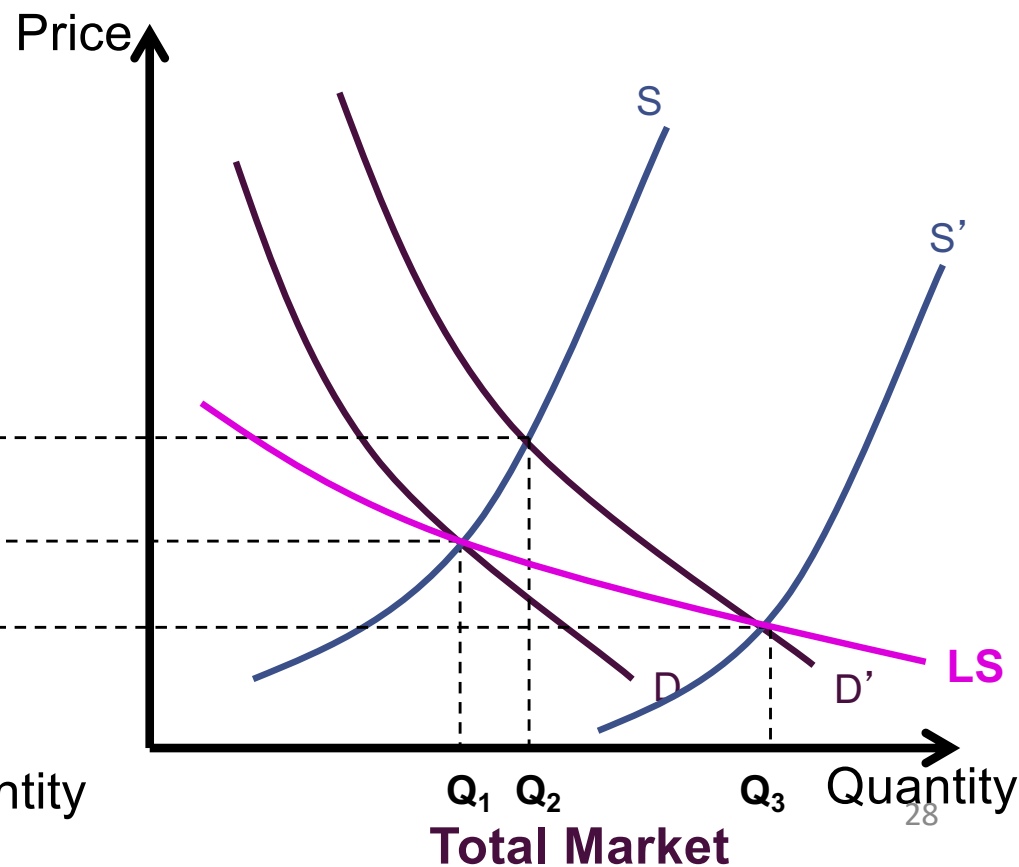
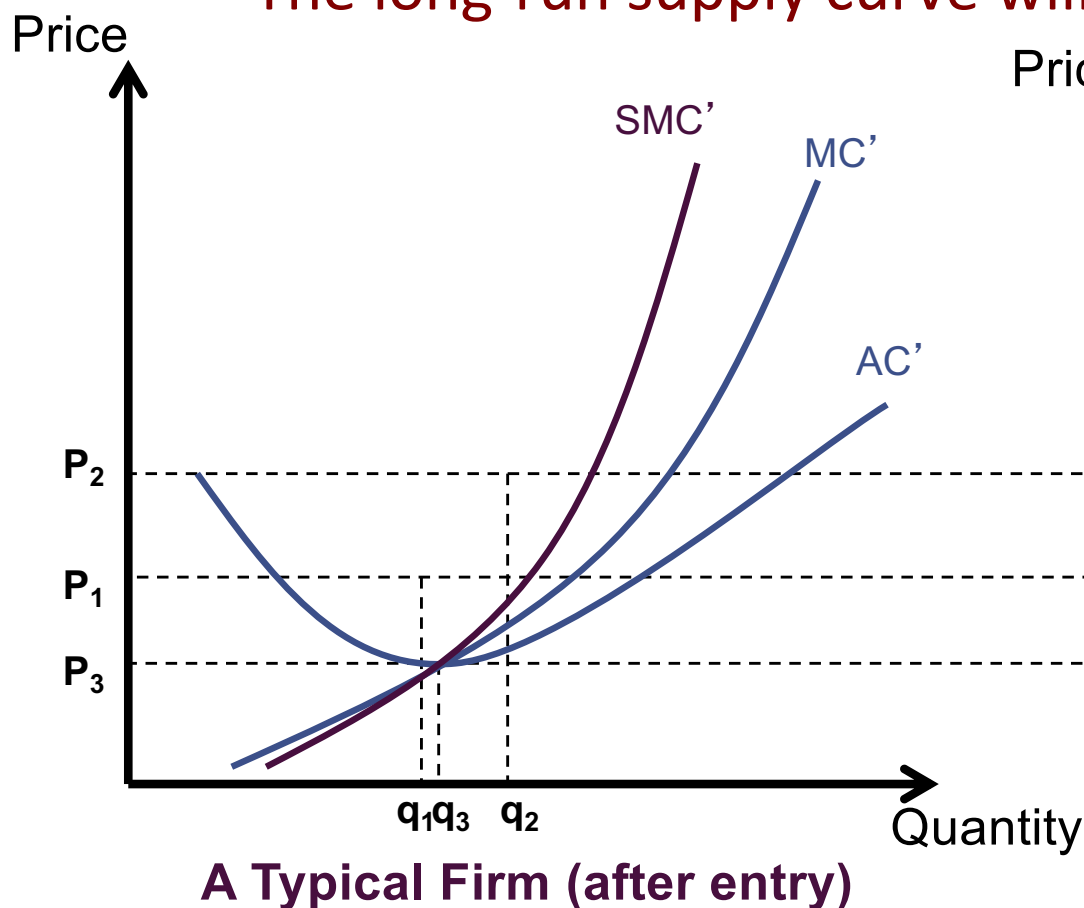
What happens if the entry of new firms reduces the costs in an industry?



4. Long-run Equilibrium: Decreasing Cost Industry

What happens if the entry of new firms reduces the costs in an industry?

The long-run supply curve will be downward-sloping!



4. Long-run Equilibrium:

Long-run Elasticity of Supply

$$e_{LS,P} = \frac{\partial Q_{LS}}{\partial P} \cdot \frac{P}{Q_{LS}}$$

Elasticity may be positive or negative depending on whether the industry exhibits increasing or decreasing costs.

Why is this elasticity important?

Comparative Statics Analysis of Long-run Equilibrium

Producer Theory

5. Comparative statics analysis of long-run equilibrium

Shifts in demand: Since the LR supply curve for a constant cost industry is infinitely elastic, shifts in demand are easy to analyse:

- In the initial equilibrium, the number of firms (n_0) is given by:

$$n_0 = \frac{Q_0}{q^*}$$

- A shift in demand will change the equilibrium output level to Q_1 , so the new equilibrium number of firms will be:

$$n_1 = \frac{Q_1}{q^*} \rightarrow n_1 - n_0 = \frac{Q_1 - Q_0}{q^*}$$

5. Comparative statics analysis of long-run equilibrium

Changes in the input costs

- Even in the case of a constant cost industry, it isn't very easy to analyse the effect of an increase in an input price.
- An increase in an input price may increase the minimum average cost $\rightarrow$ it may also change the minimum average cost OUTPUT LEVEL for the typical firm.

$$n_1 - n_0 = \frac{Q_1}{q_1^*} - \frac{Q_0}{q_0^*}$$

5. Comparative statics analysis of long-run equilibrium

Effects of an increase in fixed costs

- AC will increase
- MC must also rise (since condition $AC=MC$ still holds)
- the long-run equilibrium price will rise
- The size of the typical firm will also increase (since MC rises, the output must rise)

Increases in other types of input costs may, however, have more complex effects.

Activity 3

Remembering activity 1, a rise in costs for bicycle frame makers will alter the equilibrium, but the precise effect on market structure will depend on how costs increase.

If a rise in shop rents causes the typical frame maker's costs to increase to:

$$C(q) = q^3 - 20q^2 + 100q + 11616$$

$$Q_D = 2500 - 3P$$

- a) Determine the long-run equilibrium in this market (P^* , Q^* , q_i .)
- b) How many firms are in the market?

Producer Surplus in the Long-run

Producer Theory

6. Producer Surplus in the long-run

In the **short-run** producer's surplus represents the return to a firm's owners in excess of what would be earned if output were zero → sum of short-run profits + short-run fixed costs.

But, in **long-run equilibrium**, profits are zero, and there are no fixed costs → Owners of firms are indifferent about whether they are in a particular market.

In the constant cost case, input prices are assumed to be independent of the production level.

In the increasing cost case, entry will bid up some input prices.

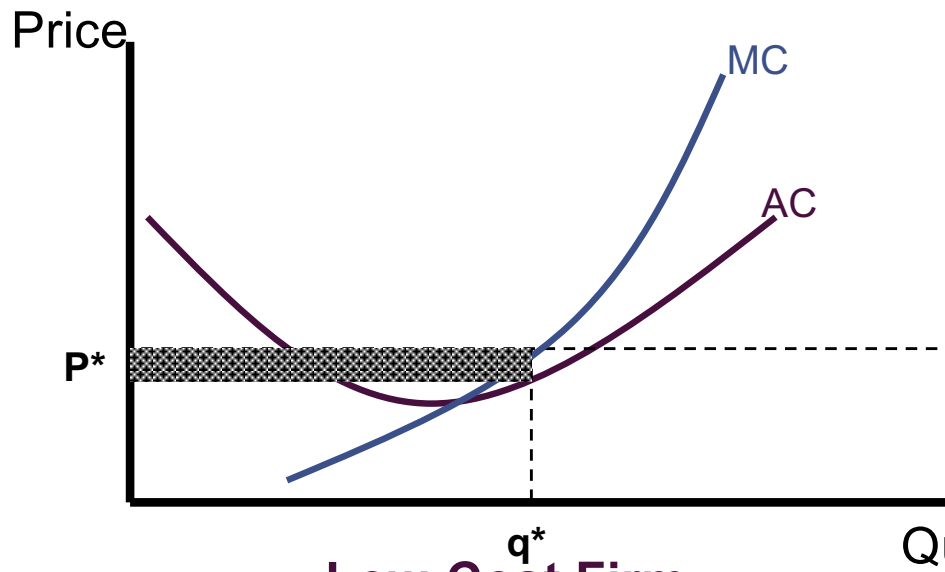
6. Producer Surplus in the long-run

Definition:

*Producer surplus is the **extra return** that producers make by making transactions at the market price over and above what they should earn if nothing were produced → size of the area below the market price and above the supply curve*

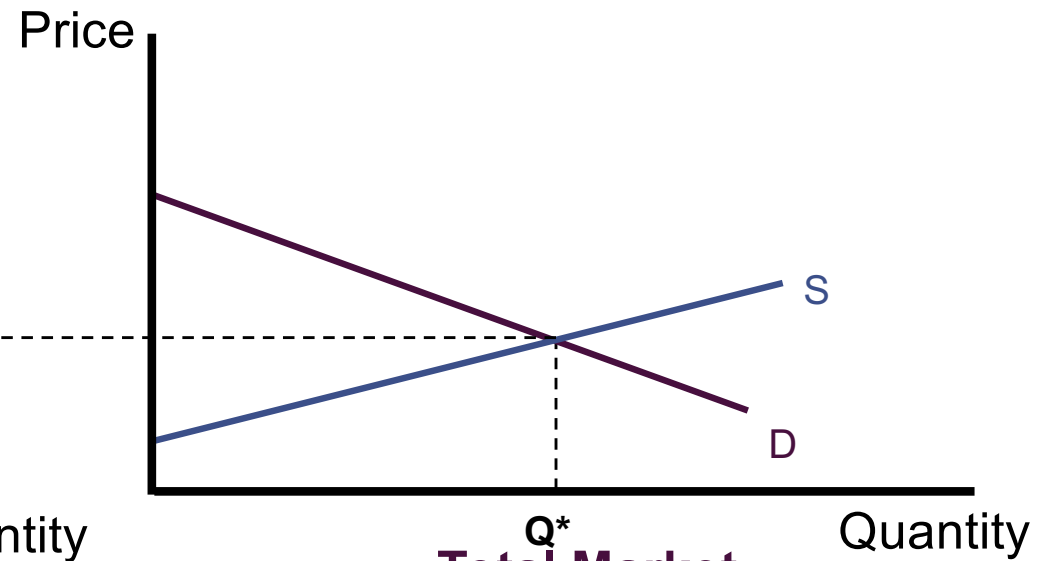
THINK: How is the producer's surplus in the constant cost industry case?

6. Producer Surplus in the long-run: Ricardian rent



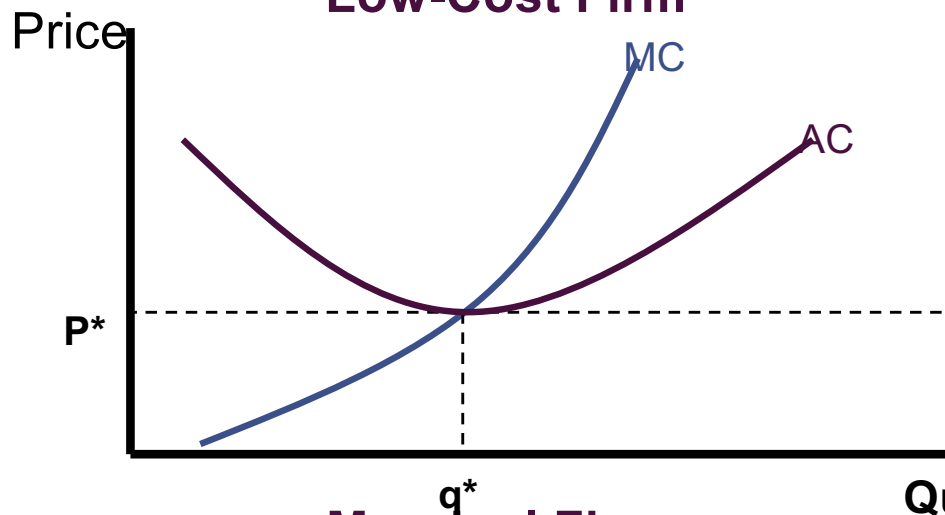
Low-Cost Firm

Quantity



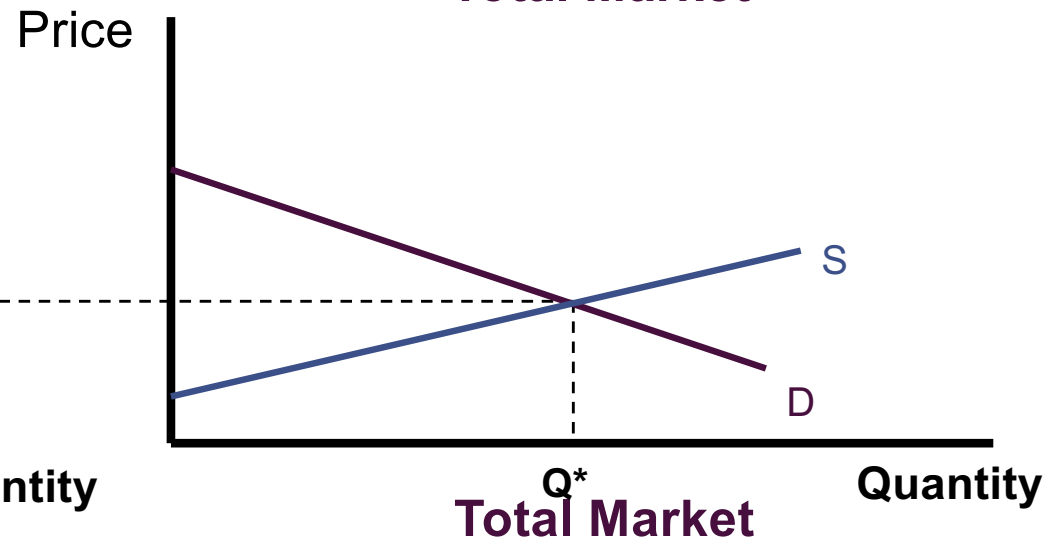
Total Market

Quantity



Marginal Firm

Quantity



Total Market

Quantity

6. Producer Surplus in the long-run

Capitalisation of rents: those LR profits for the low-cost firms should be reflected in the prices for the unique resources owned by those firms → these prices reflect the present value of all future profits → profits are said to be “capitalised” into input prices

Any scarce input will obtain rents in the form of earning a higher return than would be obtained if industry output were zero.

Economic Efficiency and Welfare Analysis

Producer Theory

7. Economic Efficiency and Welfare Analysis

Long-run competitive equilibrium may have the property of allocating resources *efficiently*

Concepts that we need to remember

- Consumer surplus
- Producer surplus

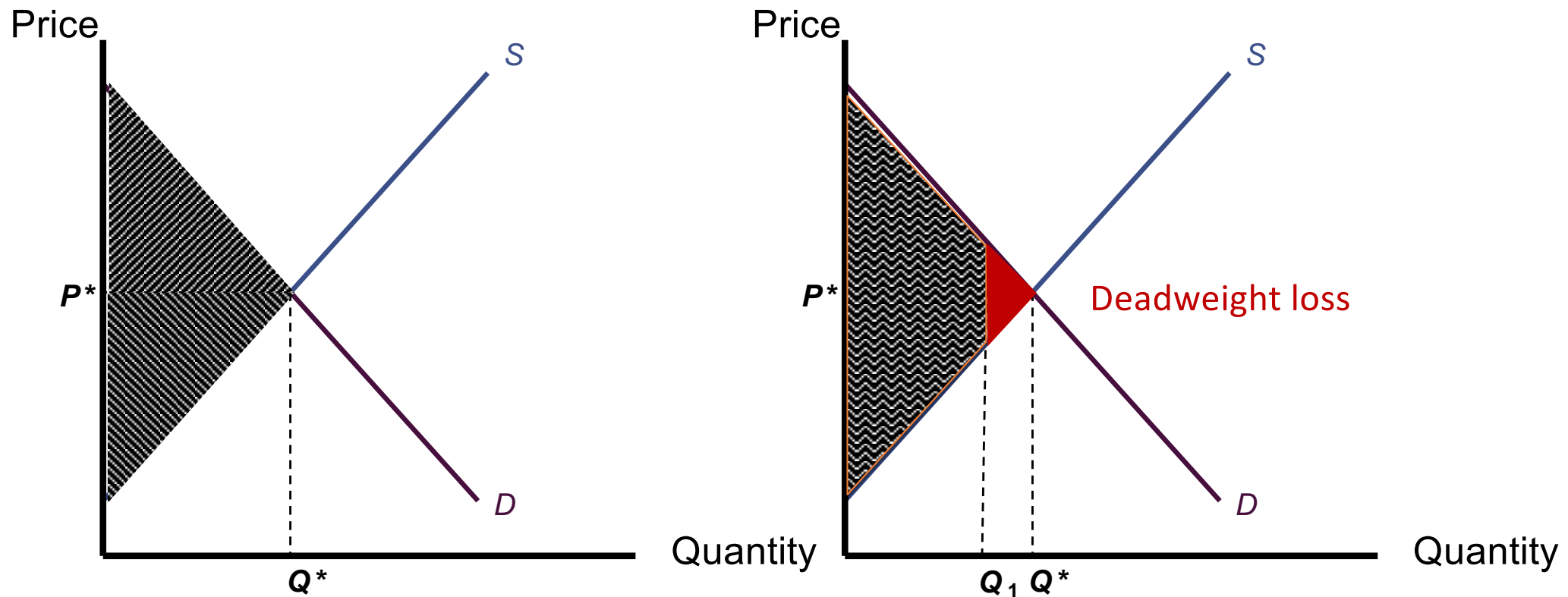
The area between the demand curve and supply curve represents the sum of consumer and producer surplus → represents the total additional value obtained by market participants

What happens at the competitive market equilibrium?

This area is MAXIMIZED!

7. Economic Efficiency and Welfare Analysis

Graphic proof



7. Economic Efficiency and Welfare Analysis

Mathematical proof

Consumer surplus + Producer surplus

$$= [U(Q) - PQ] + \left[PQ - \int_0^Q P(Q) dQ \right]$$

$$= U(Q) - \int_0^Q P(Q) dQ$$

In the LR equilibrium $\rightarrow P(Q) = AC = MC$

If we maximize the above equation with respect to Q yields:

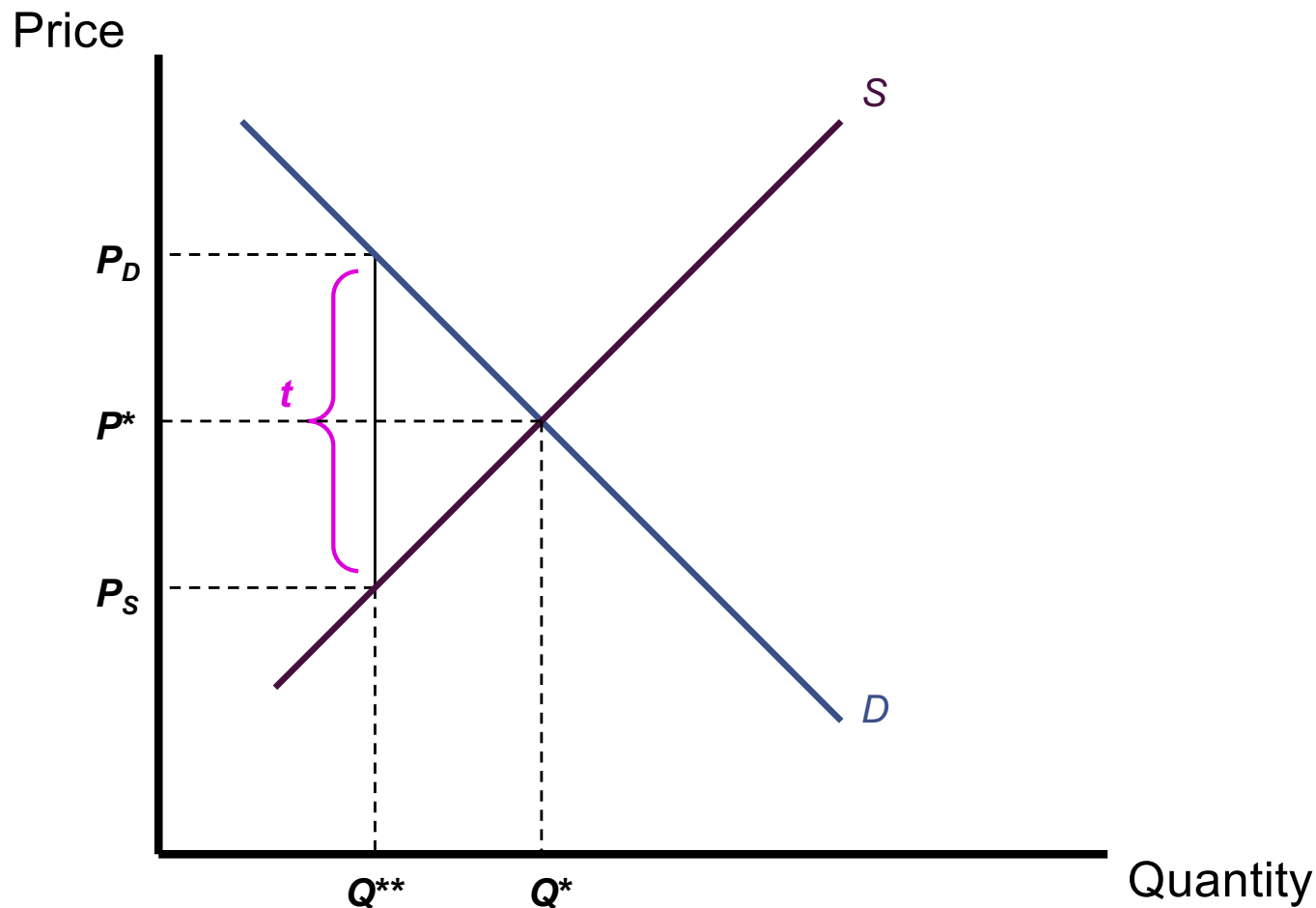
$$U'(Q) = P(Q)$$

$$\rightarrow U'(Q) = P(Q) = AC = MC$$

$\rightarrow$ Maximization occurs when the marginal value of Q to the representative consumer is equal to market price

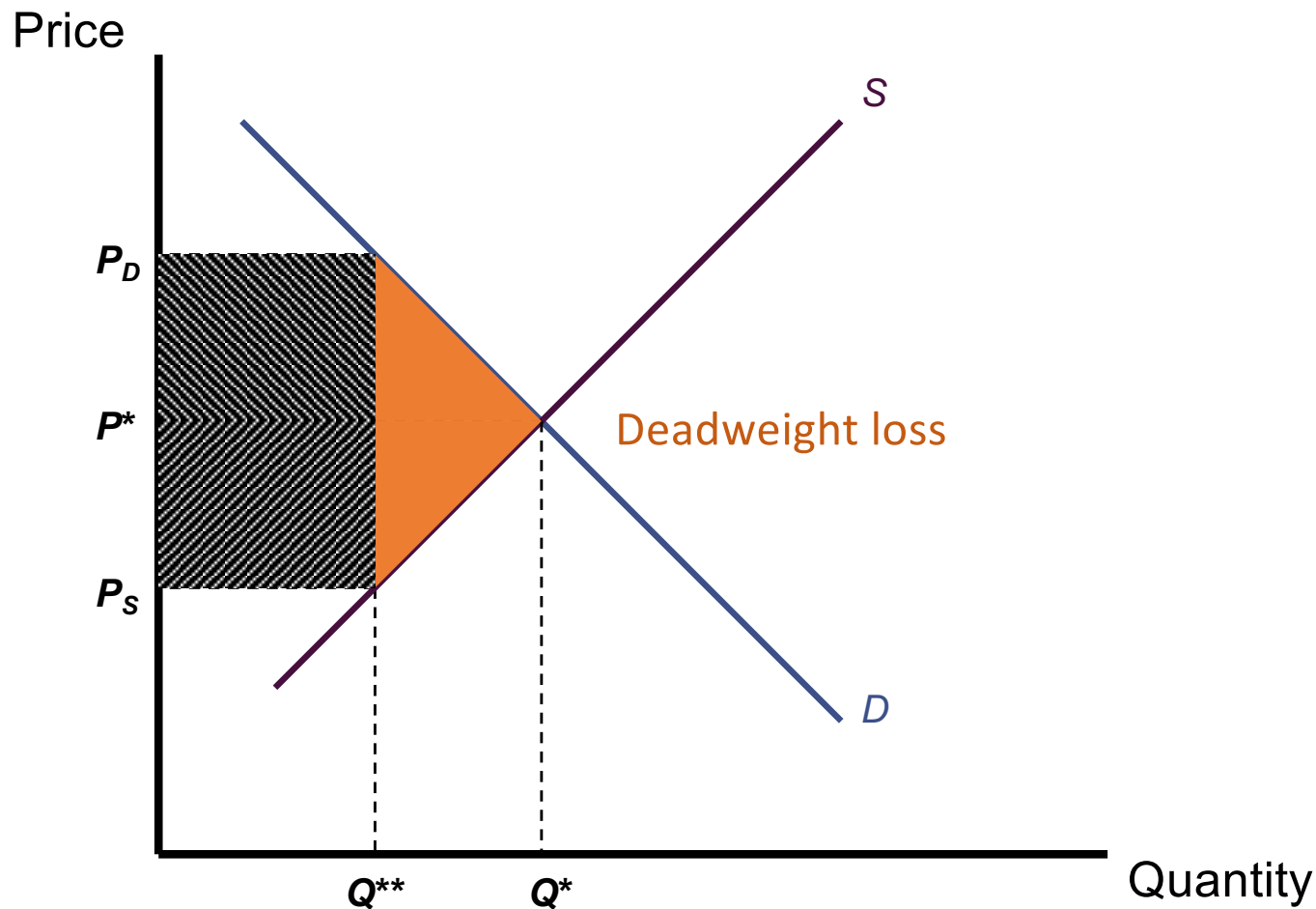
7. Economic Efficiency and Welfare Analysis

Example: Tax incidence analysis $\rightarrow P_D - P_S = t$ (per-unit tax)



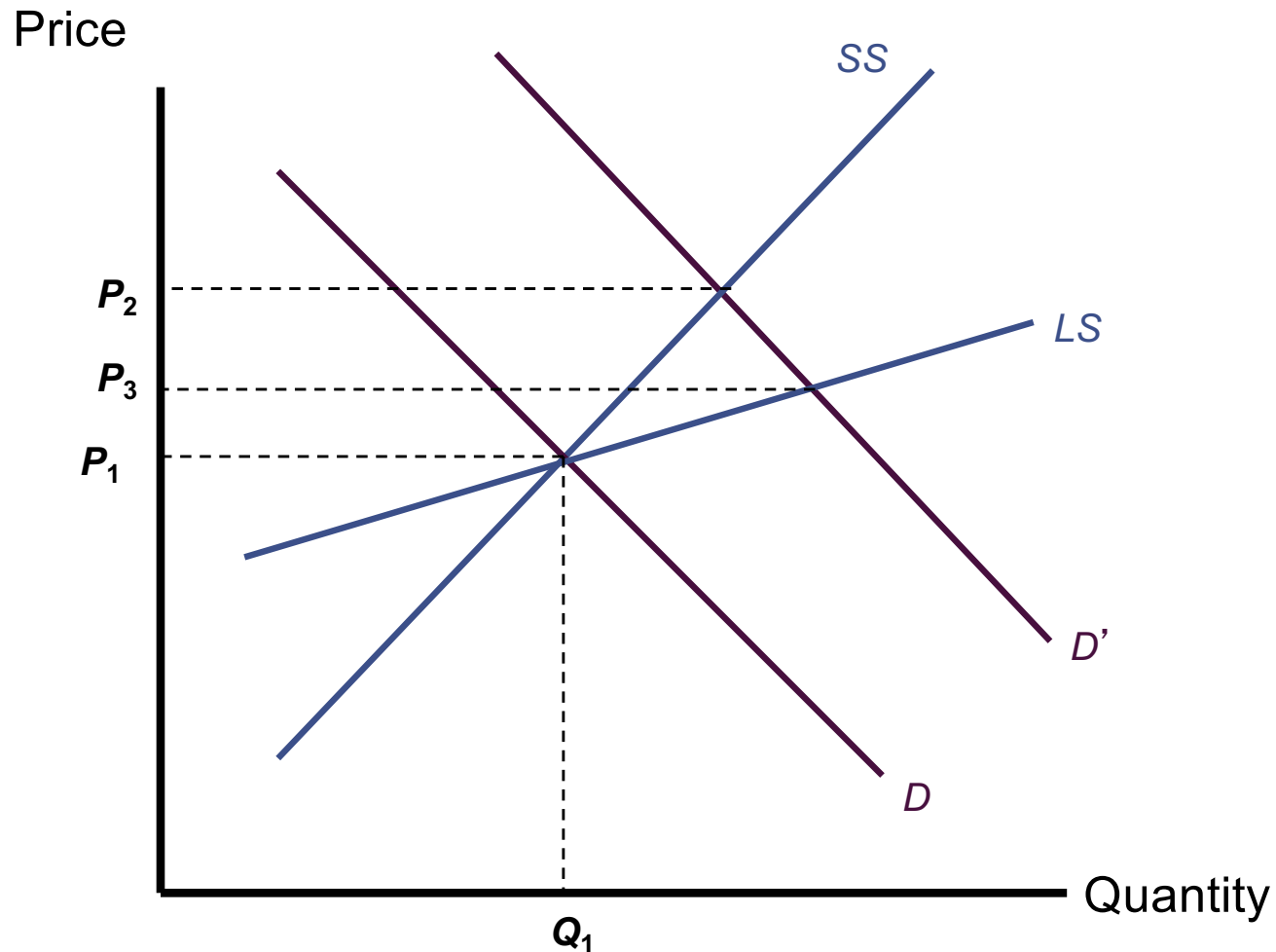
7. Economic Efficiency and Welfare Analysis

Example: Tax incidence analysis $\rightarrow P_D - P_S = t$ (per-unit tax)



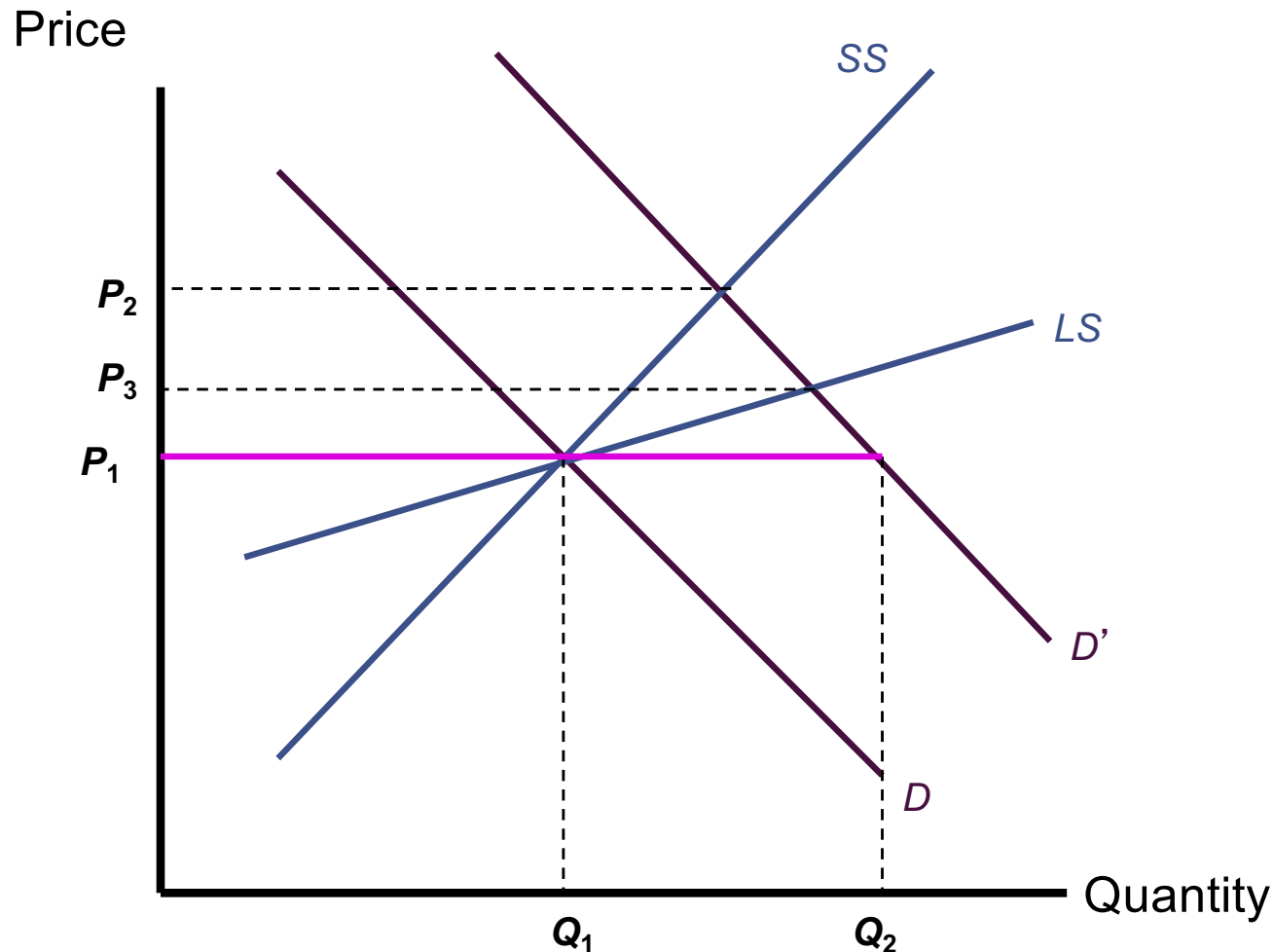
7. Economic Efficiency and Welfare Analysis

Example: Price controls and shortages



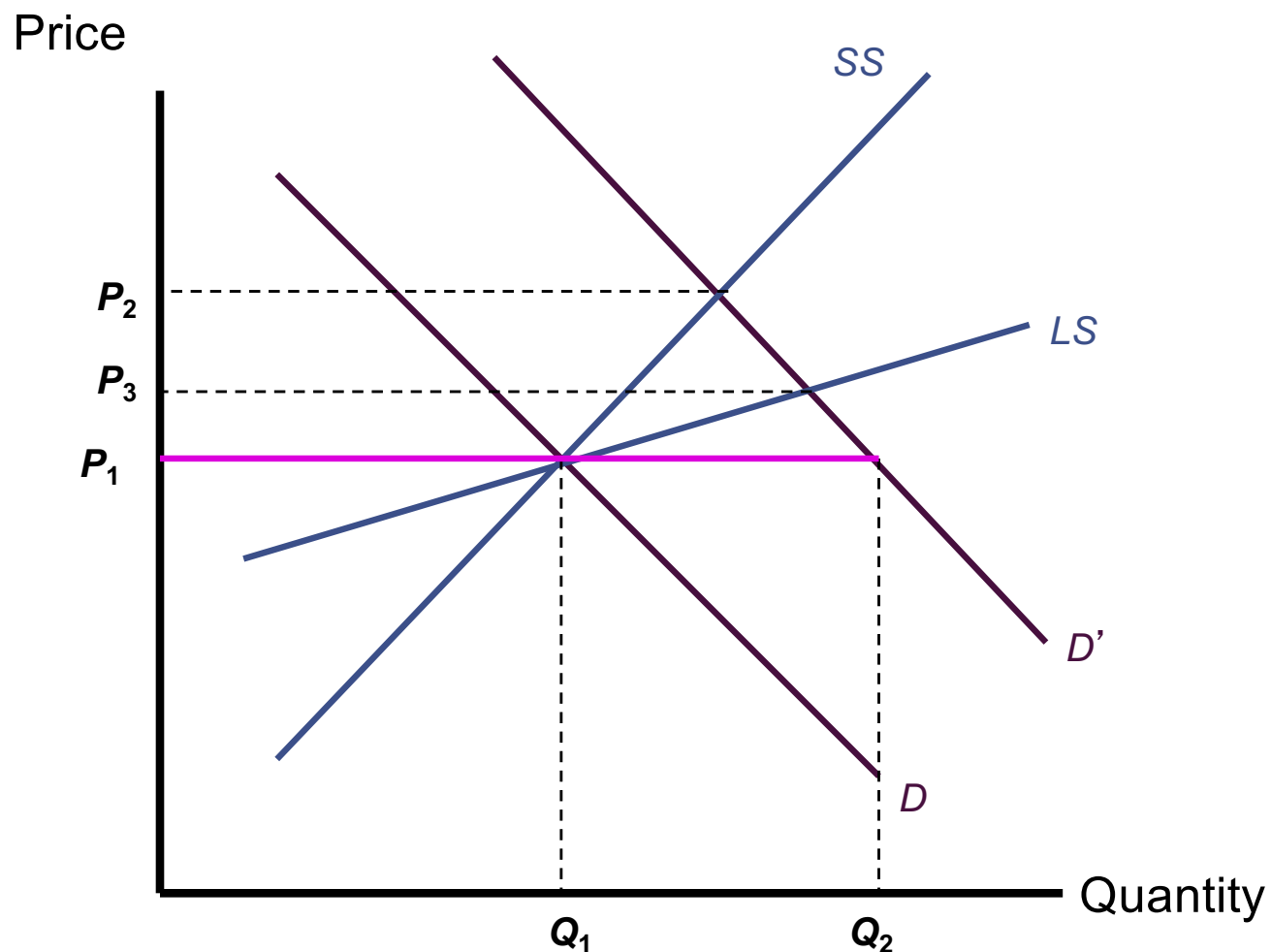
7. Economic Efficiency and Welfare Analysis

Example: Price controls and shortages



Activity 4

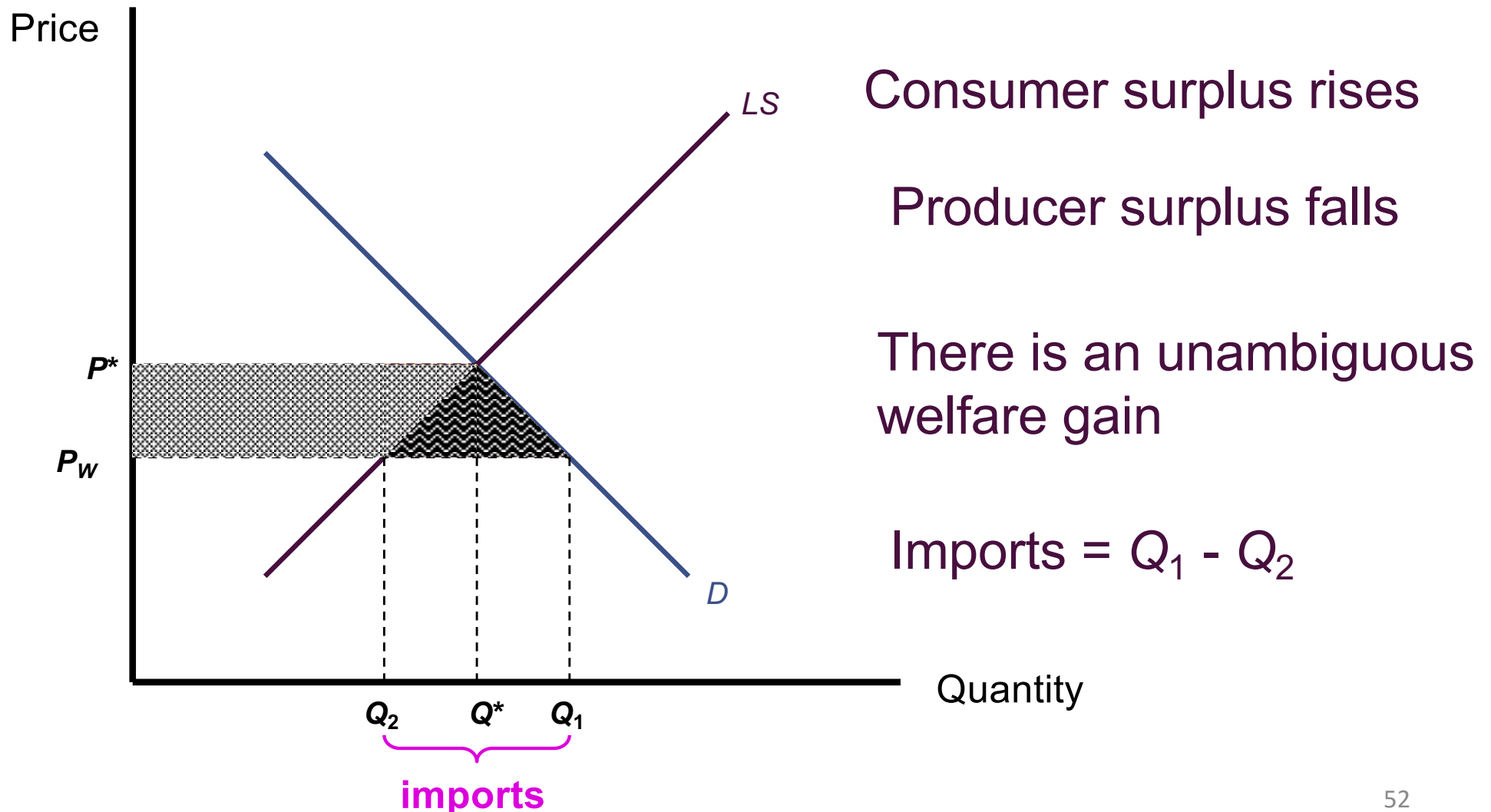
Analyze gains and losses by the use of a ceiling price (in this case).



Identify who gains, who losses, and how the welfare changes (we are better or worse?)

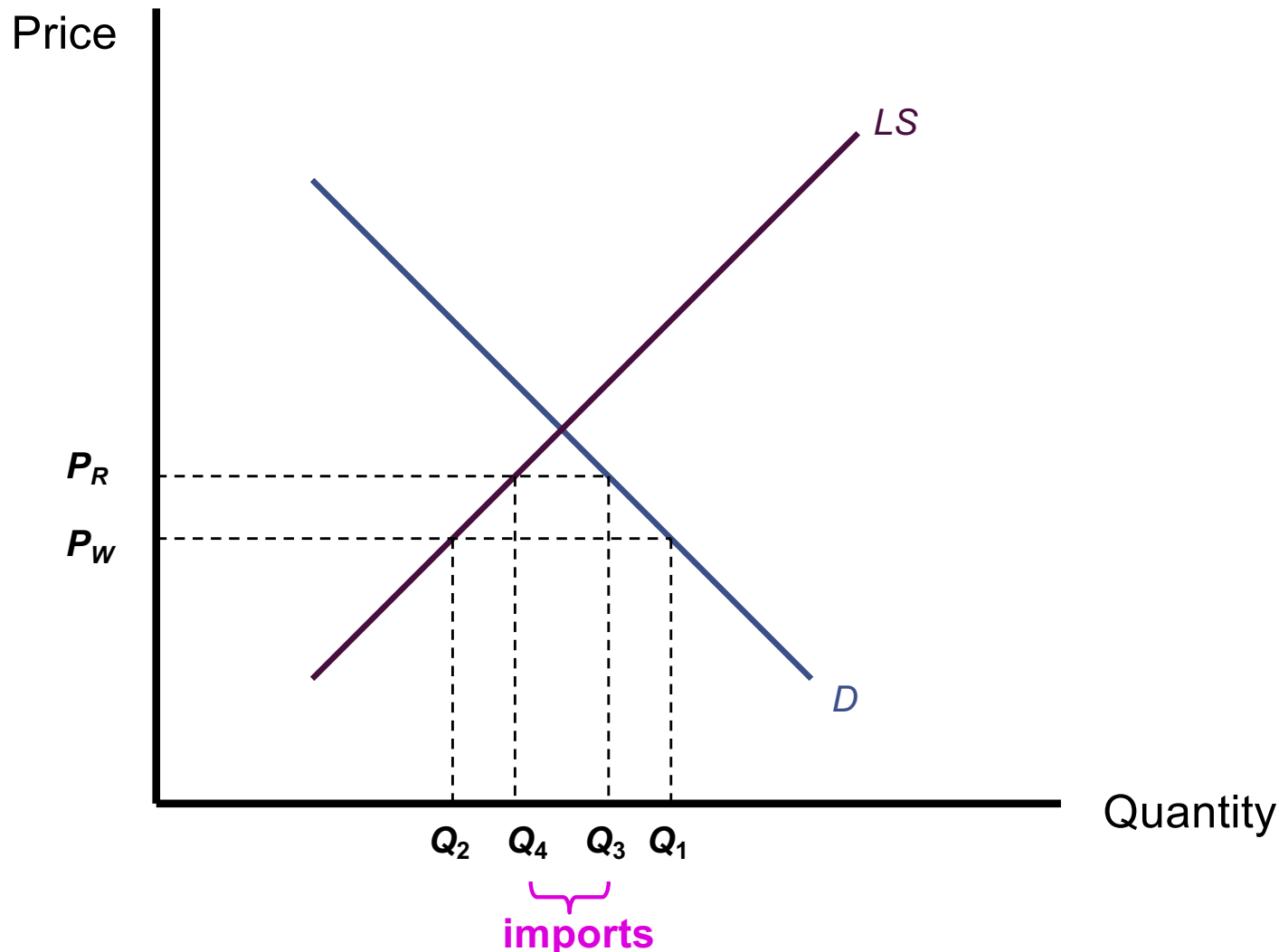
7. Economic Efficiency and Welfare Analysis

Example: Gains from international trade



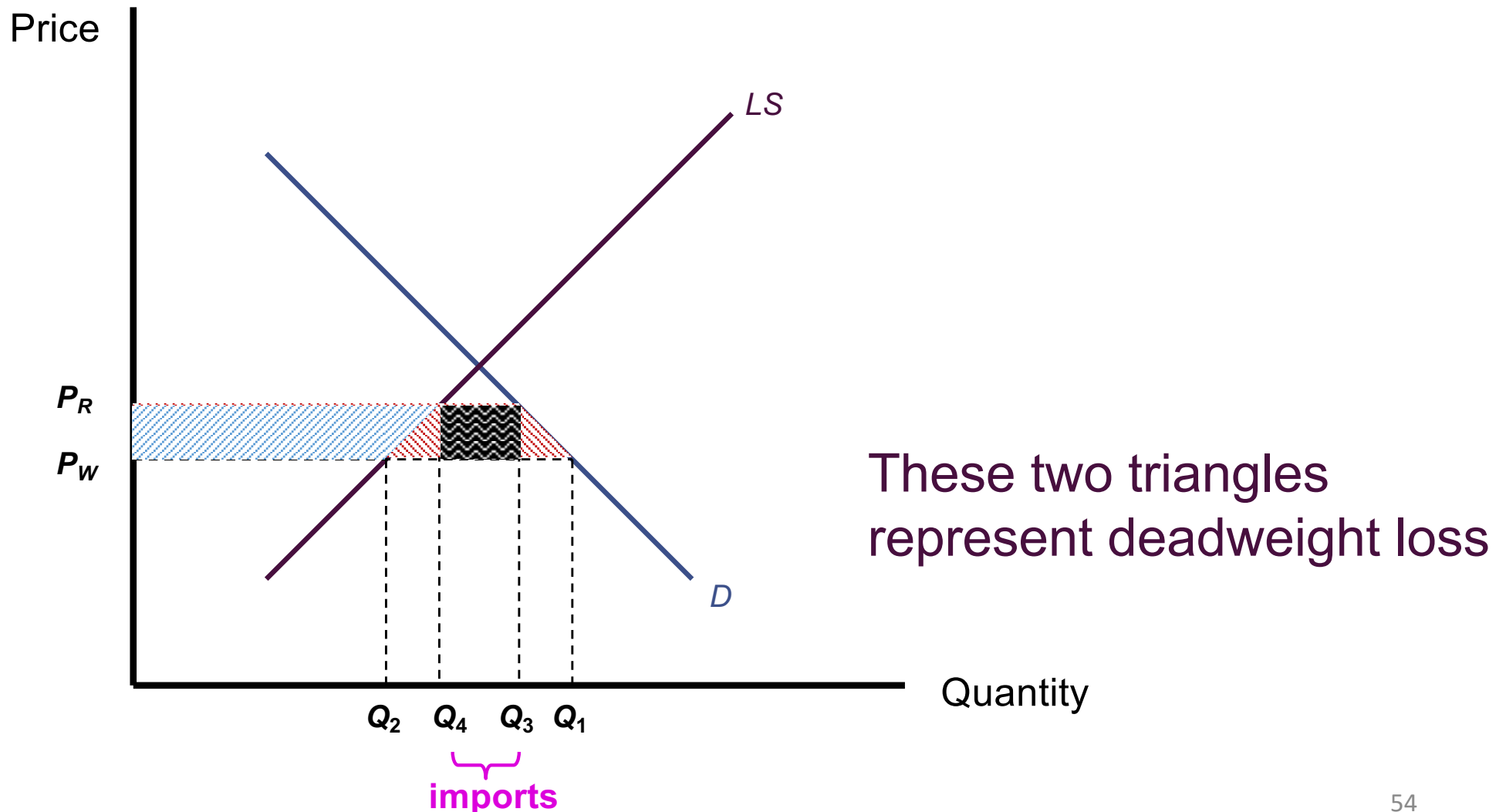
7. Economic Efficiency and Welfare Analysis

Example: Tariff protection and the politics of trade: $P_R = P_W + t$



7. Economic Efficiency and Welfare Analysis

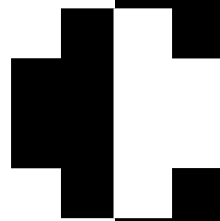
Example: Tariff protection and the politics of trade: $P_R = P_W + t$



7. Economic Efficiency and Welfare Analysis

Please review chapter 12 (The Partial Equilibrium Competitive Model) the mathematical models for these examples, and how to calculate the deadweight loss with the elasticity in the guide book Nicholson & Snyder

**Consumer
Theory**



**Producer
Theory**

Welfare Economics and Competitive Markets

Market Failures

Game Theory

Individual Activity

In a piece of paper write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it, in your own words.

If you have any **questions** about we discussed today, write it down too. We will go through them at the beginning of next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.